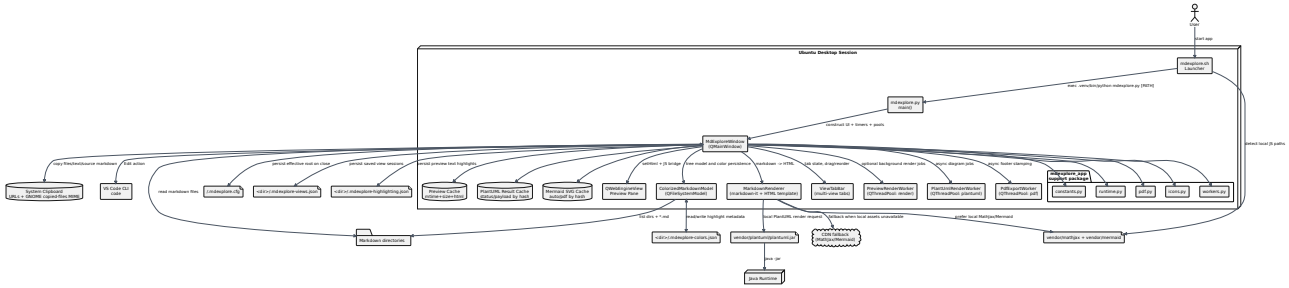
mdexplore UML

This document provides a comprehensive PlantUML view of the current `mdexplore` implementation. All diagrams are embedded so they can be rendered directly by mdexplore (or any Markdown viewer with PlantUML support).

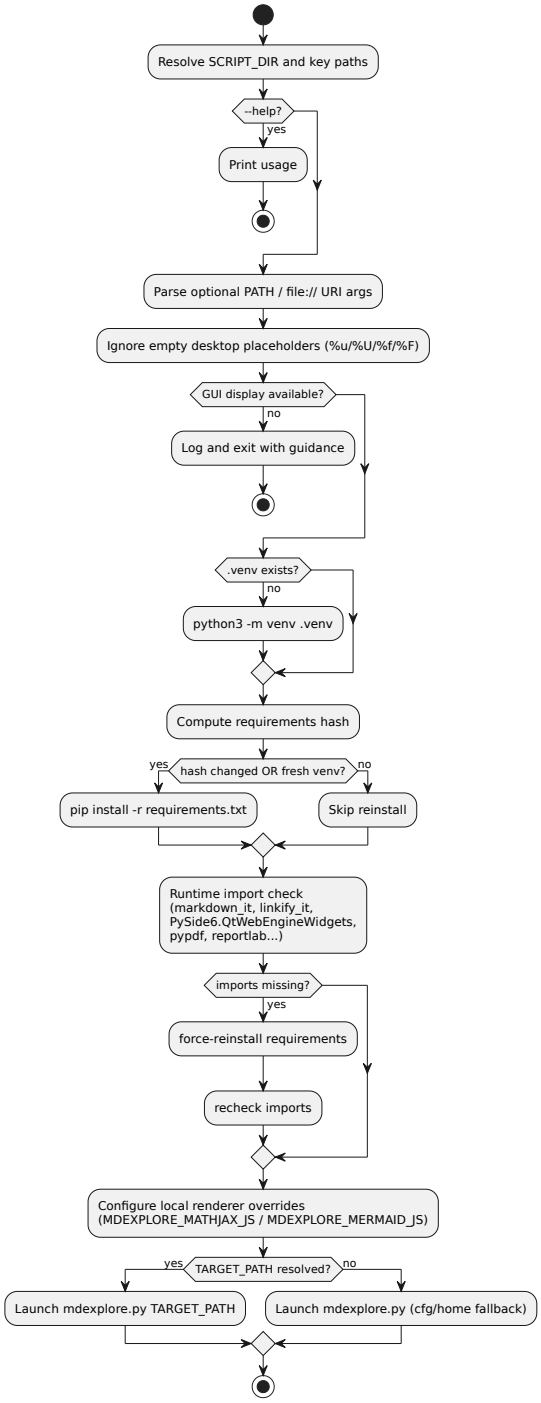
Detailed render/caching forks (GUI vs PDF, JS Mermaid vs Rust Mermaid, cache mode ownership, and restore behavior) are intentionally documented in `RENDER-PATHS.md`. This UML file keeps those areas abstracted at system/class boundaries to avoid duplicating low-level render logic across two docs.

Low-risk modularization note: support code now lives in `mdexplore_app/` (`constants.py`, `runtime.py`, `pdf.py`, `icons.py`, `workers.py`), while the main orchestration, renderer/template logic, and UI state machine remain in `mdexplore.py`. The diagrams below show that split at subsystem boundaries rather than expanding every extracted helper inline.

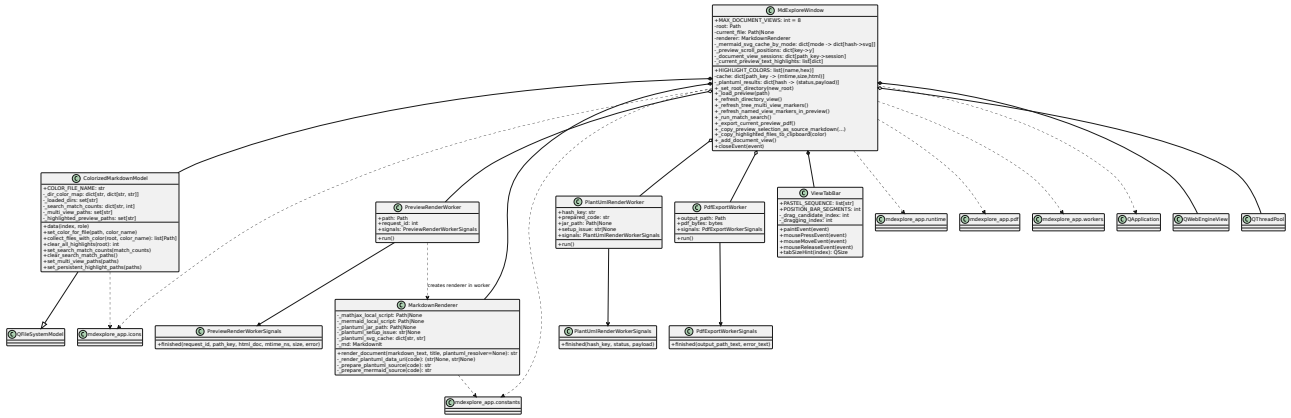
1. System Architecture


user.app
Ubuntu Desktop Session
mdexplore.sh
+.bashrc/.bif
exec -setsid/usr/bin/ mdexplore.py [PATH]
mdexplore.py
main()
construct UI + timers + pools
MdExploreWindow
(QMainWindow)
copy files/text/source markdown
trash action
persist effective root on close
persist saved view sessions
persist preview text highlights
restore + JS bridge
tree model and color persistence
markdown -> HTML
link state, drag/render
optional background render jobs
async diagram jobs
async footer stamping
mdexplore_app
module bundle
System Clipboard
(URL + GNOME copied files MIME)
VS Code CLI
code
read markdown files
.mdexplore.cfg
~/.cd/<>/.mdexplore-views.pickle
~/.cd/<>/.mdexplore-highlighting.pickle
Preview Cache
(active + LLM-TOC)
PlantUML Result Cache
(shared/pulled by hash)
Mermaid SVG Cache
(shared/ by hash)
QWebEngineView
Preview Pane
QWebChannelBridge
(QFileSystemModel)
MarkdownRenderer
(markdown-it + HTML templates)
ViewTabBar
(multi-view tabs)
PreviewRenderWorker
(QThreadPool: render)
PlantUmlRenderWorker
(QThreadPool: plantuml)
MdExportWorker
(QThreadPool: pdf)
constants.py
outline.py
pdf.py
icons.py
workers.py
Markdown directories
~/.cd/<>/.mdexplore-views.pickle
~/.cd/<>/.mdexplore-highlighting.pickle
~/.cd/<>/.mdexplore-colors.pickle
~/.cd/<>/.mdexplore-cache/
~/.cd/<>/.mdexplore-cache/
~/.cd/<>/.mdexplore-cache/

2. Launcher Runtime Activity (`mdexplore.sh`)

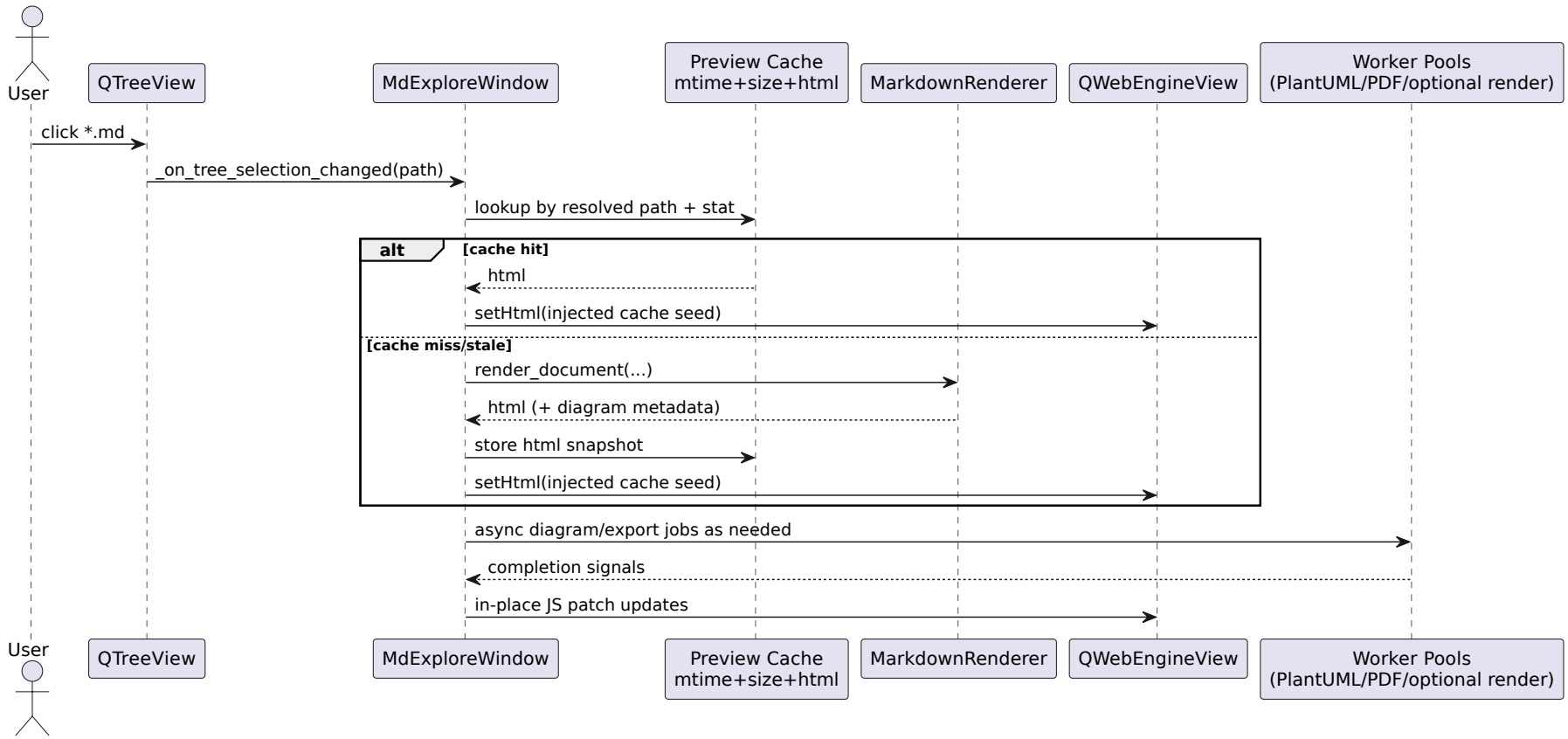


3. Core Class Diagram



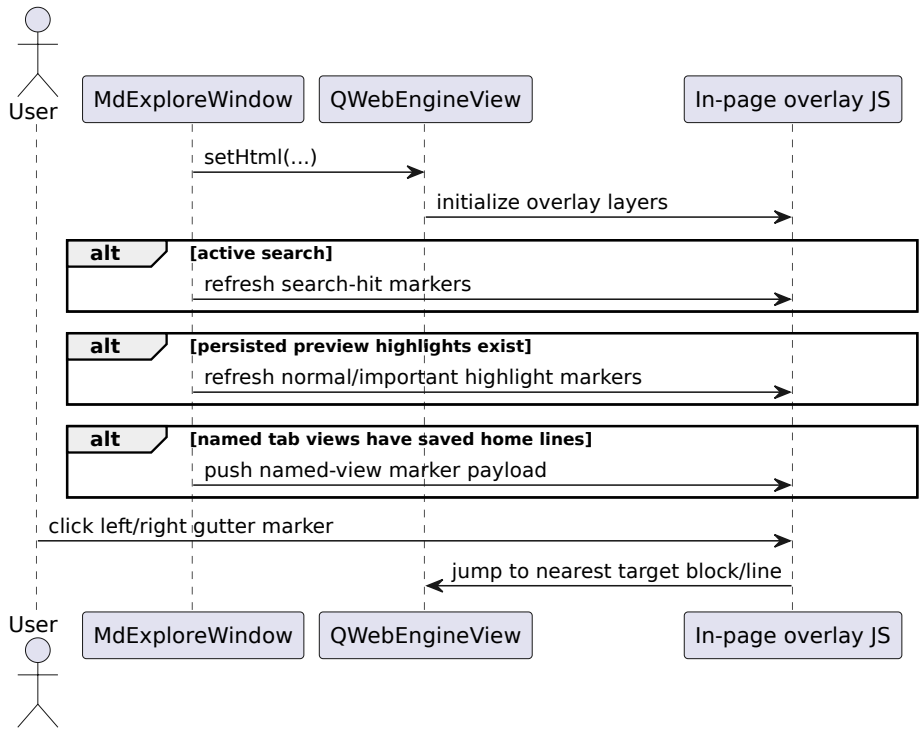
4. Preview Pipeline (Abstracted)

The detailed render/cache branches are covered in `RENDER-PATHS.md`. This section intentionally keeps an architectural boundary view only.

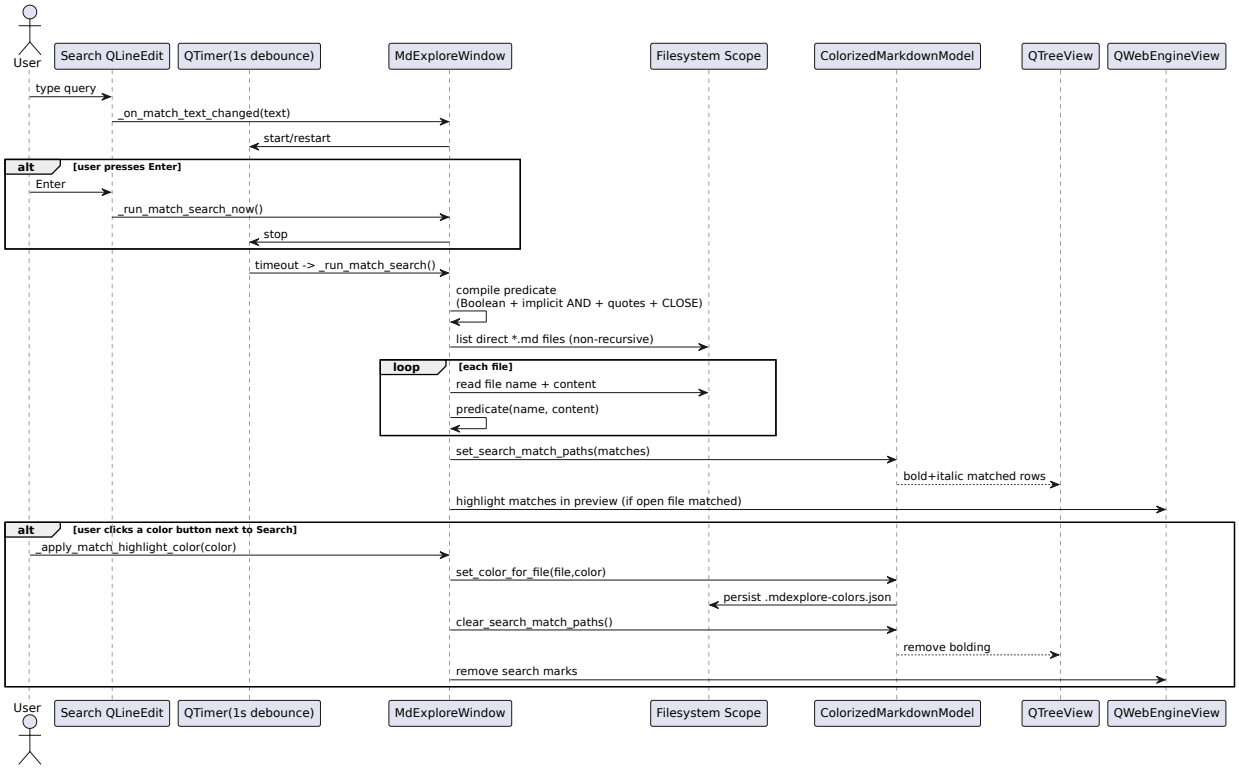


4.1 Preview Navigation Overlays

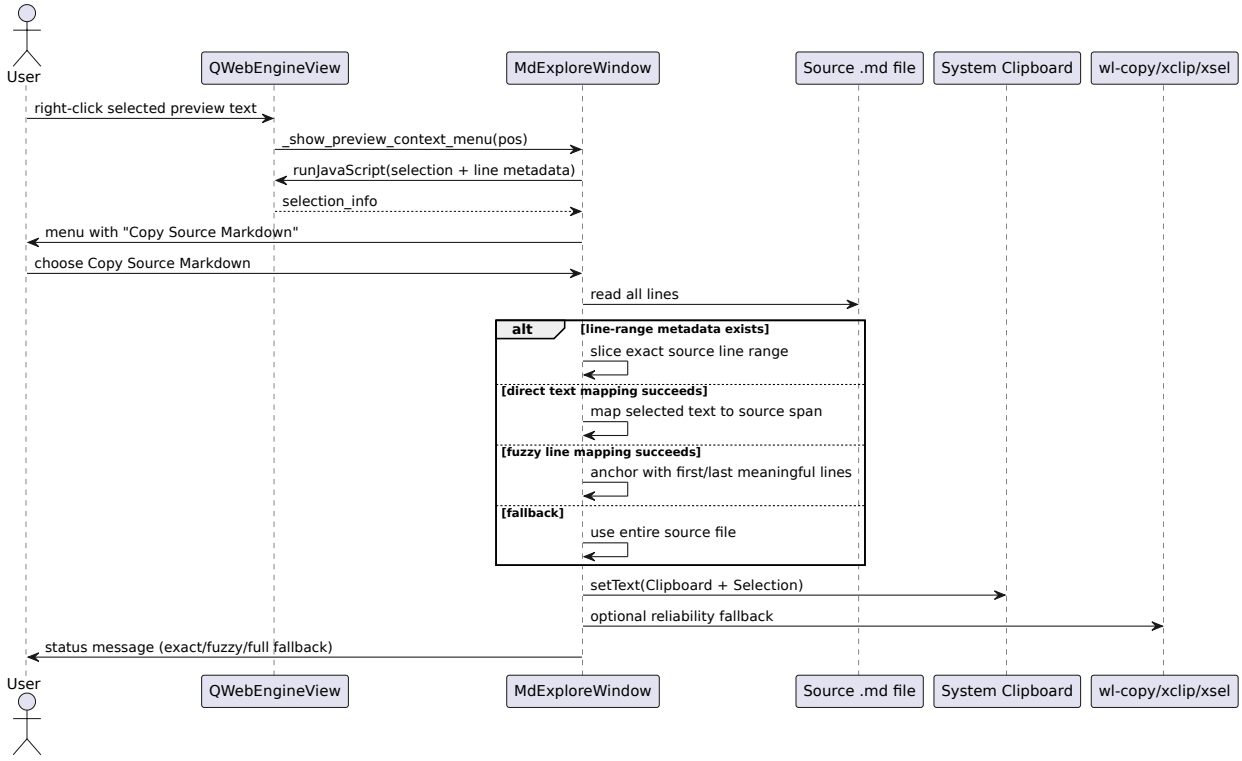
These are intentionally shown here only as boundary-level UI overlays. The exact DOM/CSS/JS mechanics remain in `RENDER-PATHS.md`.



5. Search + Highlight + Apply Color Flow

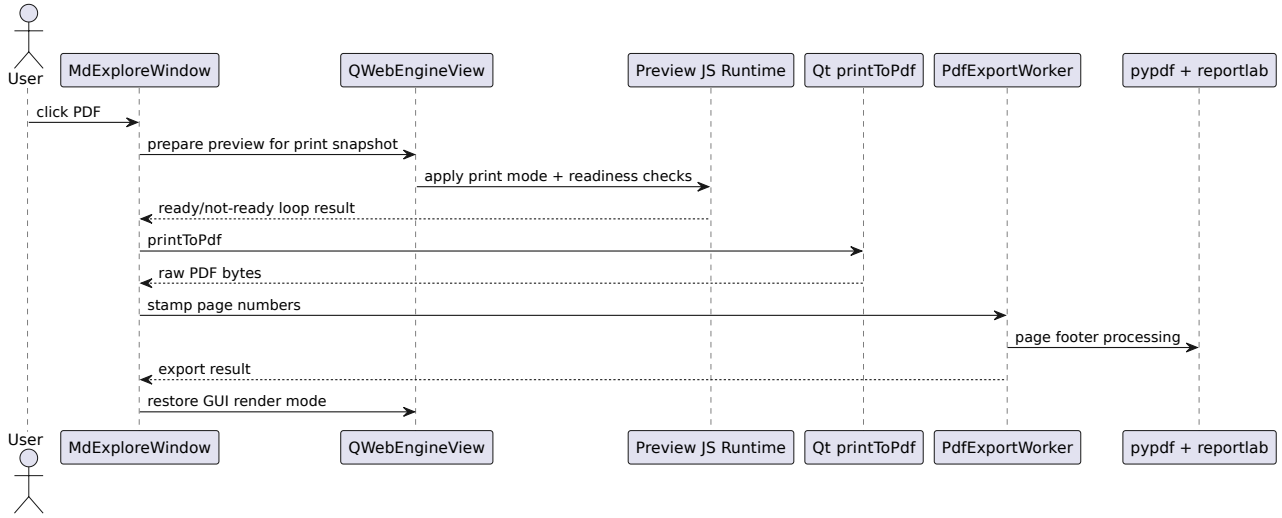


6. Preview Context Menu: Copy Source Markdown

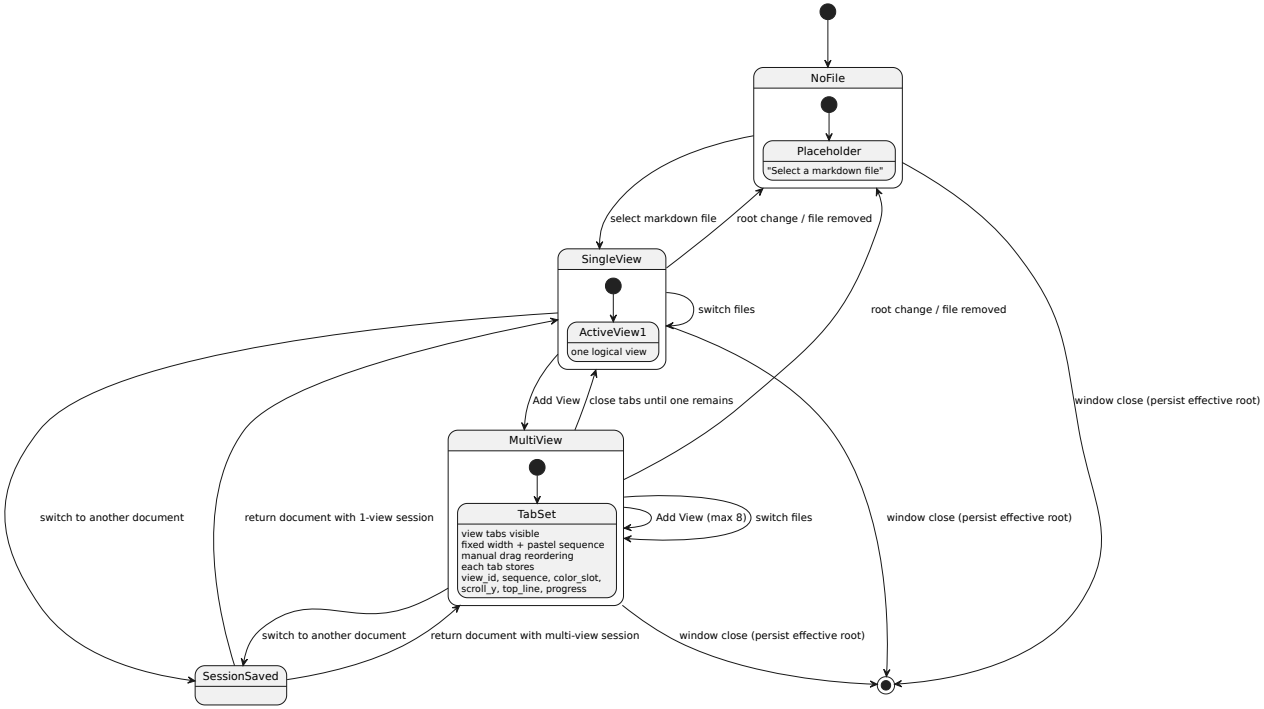


7. PDF Export Pipeline (Abstracted)

Detailed PDF mode branching (JS Mermaid grayscale path vs Rust Mermaid default-themed PDF cache path) is documented in `RENDER-PATHS.md`.



8. Document View/Tab Lifecycle



Notes

- Diagrams are based on current code in `mdexplore.py` and `mdexplore.sh`.
- Render/caching branch internals are intentionally abstracted here and documented in `RENDER-PATHS.md` to keep a single authoritative deep map.
- Preview-only zoom (`Ctrl++` , `Ctrl+-` , `Ctrl+0`) is also intentionally kept out of the UML internals here because it is a `QWebEngineView` scale adjustment, not a separate renderer/cache branch.
- Preview gutter overlays (search-hit markers, persistent-highlight markers, named-view home markers) are similarly abstracted here and described in more detail in `RENDER-PATHS.md`, because they are post-render navigation aids rather than renderer/cache forks.
- Worker/threadpool usage is intentionally separated by concern:
  - render pool (preview HTML jobs),
  - PlantUML pool (diagram jobs),
  - PDF pool (post-processing/stamping).
- PlantUML rendering is non-blocking in UI flow: placeholders are rendered first, then patched in place as jobs finish.
- TODO (known issue): diagram zoom/pan restore (Mermaid and PlantUML) is not yet consistently reliable when leaving a document and returning in the same app run.